One Result, Two Claims

Unit 3 · Topic 3.13 · THE PROBLEM

Suppose a utility has 500,000 online accounts. It randomly selects 10,000, randomly assigns 5,000 to message A and 5,000 to B, and records on-time payment. The pre-data question asks whether the population proportions differ at $\alpha = 0.05$; deployment requires A to improve payment by at least 5 percentage points.

Rowan: "I rejected equality, so A has proved an important improvement of at least 5 points. Deploy A."

Salma: "Rejecting equality supports a nonzero difference, but does not automatically establish the separate 5-point requirement."

Conditions have been verified: random selection and assignment; 10,000 $\leq$ 50,000; pooled expected counts 3,000 and 2,000 in each group.

On-time payment

Text	n	Paid	Other
A	5000	3050	1950
B	5000	2950	2050

FIRST TAKE — one sentence before you work the parts

The observed improvement is 2 percentage points and the deployment target is 5. Is that enough to deploy? Give one reason.

DOK 1 **(a)** Calculate $\hat{p}_A$, $\hat{p}_B$, and the observed difference $\hat{p}_A - \hat{p}_B$.

DOK 2 **(b)** Using the verified conditions, calculate the pooled proportion, pooled SE, z, and the two-sided p-value for $H_0 : p_A - p_B = 0$.

DOK 3 **(c)** In 3–4 sentences, choose a report using the p-value, the 2-point observed gap, and the 5-point target. Explain what the test against zero establishes and why that does or does not justify deployment.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.